REPRESENTABLE FUNCTORS ARE SHEAVES FOR FPQC COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof's closing diagram chase is carried in one sentence by the author; all ingredients (uniqueness, quotient presentation) are stated in the span.

Allowed standard prerequisites (editorial).

- fpqc sheaf property for schemes. Stacks Project, Properties of Spaces, Proposition spaces-properties-proposition-sheaf-fpqc. Supplies the equalizer property for the fpqc covering $\{U_i \to U\}$ of schemes.
- fpqc coverings by base change and recognition. Stacks Project, Topologies on Spaces, Lemmas spaces-topologies-lemma-fpqc and spaces-topologies-lemma-recognize-fpqc-covering. Shows $\{U_i \to U\}$ is an fpqc covering.
- Algebraic spaces as sheaf quotients. Stacks Project, Descent on Spaces, discussion of $T = U/(U \times_T U)$ in the chapter text. Supplies the descent of a from U to T.

Source reference. Original reference 03MQ: sources/stacks-chow/tags.txt, source lines 4758–4758. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0APL: sources/stacks-chow/tags.txt, source lines 13479–13479. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03MS: sources/stacks-chow/tags.txt, source lines 4760–4760. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $\{T_i \to T\}$ be an fpqc covering of algebraic spaces over S, see Topologies on Spaces, Definition 03MQ. Then given an algebraic space B over S the sequence*

$$\mathrm{Mor}_S(T, B) \longrightarrow \prod_i \mathrm{Mor}_S(T_i, B) \rightrightarrows \prod_{i,j} \mathrm{Mor}_S(T_i \times_T T_j, B)$$

is an equalizer diagram. In other words, every representable functor on the category of algebraic spaces over S satisfies the sheaf condition for fpqc coverings.

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Proof. We know this is true if $\{T_i \to T\}$ is an fpqc covering of schemes, see Properties of Spaces, Proposition 0APL. This is the key fact and we encourage the reader to skip the rest of the proof which is formal. Choose a scheme U and a surjective étale morphism $U \to T$. Let U_i be a scheme and let $U_i \to T_i \times_T U$ be a surjective étale morphism. Then $\{U_i \to U\}$ is an fpqc covering. This follows from Topologies on Spaces, Lemmas 03MR and 03MS. By the above we have the result for $\{U_i \to U\}$.

What this means is the following: Suppose that $b_i : T_i \to B$ is a family of morphisms with $b_i \circ \mathrm{pr}_0 = b_j \circ \mathrm{pr}_1$ as morphisms $T_i \times_T T_j \to B$. Then we let $a_i : U_i \to B$ be the composition of $U_i \to T_i$ with b_i. By what was said above we find a unique morphism $a : U \to B$ such that a_i is the composition of a with $U_i \to U$. The uniqueness guarantees that $a \circ \mathrm{pr}_0 = a \circ \mathrm{pr}_1$ as morphisms $U \times_T U \to B$. Then since $T = U/(U \times_T U)$ as a sheaf, we find that a comes from a unique morphism $b : T \to B$. Chasing diagrams we find that b is the morphism we are looking for. $\square$